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An overview of 3D printing Mars habitats using in-situ resources: materials, processing, structural challenges, and prospects

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ABSTRACT

The establishment of a sustained human presence on Mars is strongly constrained by the mass, cost, and logistics associated with transporting construction materials from Earth, making in-situ resource utilization (ISRU) a central enabling strategy for surface infrastructure. Among ISRU-compatible approaches, additive manufacturing (3D printing) has emerged as a promising solution due to its compatibility with robotic deployment, efficient material usage, and geometric flexibility under extraterrestrial conditions. This review synthesizes current knowledge on Martian regolith as a construction feedstock, addressing its mineralogical, chemical, and physical characteristics, as well as the extreme environmental constraints imposed by Mars, including low atmospheric pressure, large diurnal temperature variations, high radiation flux, and limited water availability. This paper critically reviews the main 3D printing technologies and material systems proposed for Martian habitat construction, including extrusion-based alkali-activated binders, sulfur-based thermoplastic concretes, and binder-free sintering approaches driven by solar, microwave, or laser energy. Protective design strategies such as regolith overburden, hybrid inflatable-rigid architectures, and consolidated regolith shells are discussed with respect to radiation shielding, thermal buffering, and structural integrity under pressurization. By integrating experimental studies, numerical modeling, and conceptual designs, this review identifies key technological challenges and research gaps related to durability, energy efficiency, scalability, and long-term performance, and outlines critical pathways toward the realization of scalable and sustainable regolith-based habitats for long-duration human missions to Mars.

1. Introduction

Human exploration and long-term habitation of Mars represent one of the most ambitious technological challenges of the twenty-first century. Beyond propulsion and life-support systems, the feasibility of sustained human presence on Mars is fundamentally constrained by the cost, mass, and logistics of surface infrastructure. Jones [1] estimates that a first human mission to Mars would require an investment on the order of USD 300-600 billion, with a robust central estimate near USD 500 billion, based on analogies with Apollo, the Space Shuttle, and the International Space Station. These analyses consistently show that mission costs are dominated by hardware development, followed by launch and operational expenditures. Even subsystems often regarded as secondary, such as life-support, are projected to require investments of several billion dollars due to stringent reliability and redundancy requirements. Under conventional aerospace paradigms, therefore, human Mars exploration remains a half-trillion-dollar endeavor.

A central driver of this cost is the mass penalty associated with transporting materials from Earth to the Martian surface. This constraint has profoundly shaped architectural and engineering strategies for extraterrestrial construction, shifting the focus away from logistics optimization toward the elimination of terrestrial materials altogether. Within this context, In-Situ Resource Utilization (ISRU) has emerged as a foundational principle for Mars exploration, advocating the extraction and processing of local resources, primarily Martian regolith, to produce construction materials, structural components, and protective systems directly on site.

Among ISRU-enabled construction strategies, additive manufacturing (3D printing) has gained particular prominence due to its compatibility with robotic deployment and efficient use of raw materials. When coupled with regolith-derived feedstocks, additive manufacturing offers a pathway toward scalable and autonomous construction under Martian environmental constraints. A wide range of material systems and processing techniques have therefore been proposed, including extrusion-based geopolymer and sulfur concretes, thermoplastic regolith composites, and binder-free sintering approaches

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driven by solar, microwave, or laser energy (detailed in section 4). Nevertheless, the implementation of these technologies remains challenged by operational limitations such as restricted robotic dexterity, dust contamination, maintenance complexity, and the need for highly reliable autonomous systems.

At the same time, Mars presents an exceptionally hostile building environment. The planet's thin atmosphere, extreme diurnal temperature variations, absence of a global magnetic field, and high surface radiation flux impose stringent requirements on both materials and structural design [2,3]. Construction strategies must therefore address not only mechanical stability and printability, but also radiation shielding, thermal regulation, durability under cyclic thermal stress, and long-term airtightness under pressurization. These constraints have increasingly driven habitat concepts toward regolith-shielded, partially buried, or subsurface architectures, often combining lightweight deployable habitats with massive in-situ protective structures.

While additive manufacturing has emerged as a leading candidate for in-situ construction on Mars, it is not the only viable approach. Alternative strategies include the deployment of prefabricated inflatable or rigid modules, robotic assembly of modular elements, and the utilization of natural geological formations such as lava tubes. Each approach presents distinct advantages in terms of deployment speed, structural reliability, and technological readiness. Consequently, the relevance of 3D printing must be assessed within this broader context, particularly in relation to scalability, material utilization, and long-term sustainability.

Addressing these challenges requires not only advances in materials and construction technologies, but also their integration within a broader interdisciplinary framework. Mars habitat development has thus evolved into a highly interdisciplinary field that extends beyond materials and construction to encompass architectural design, human factors, life support systems, and planetary environmental sciences. Accordingly, this work focuses on the material and process aspects of regolith-based additive manufacturing for 3D-printed protective shells in inflatable habitat systems, while complementary aspects of Mars habitation - including architectural design [4,5], system-level and mission integration [6], biological constraints [7], life support systems, habitability and human factors [8], ethical and societal considerations [9], astrochemical and radiation-driven processes [10], and interdisciplinary mission feasibility [11] - are addressed in the corresponding literature.

In this context, the present review provides a critical assessment of 3D printing technologies for Mars habitat construction using in-situ resources. The paper synthesizes current knowledge on Martian regolith properties, environmental constraints, printable material systems, and additive manufacturing techniques, with particular emphasis on extrusion-based binders, sulfur-based thermoplastics, and high-temperature sintering approaches. By integrating materials science, construction engineering, and planetary environmental considerations, this review aims to clarify the technical feasibility, limitations, and developmental pathways of 3D-printed Mars habitats, highlighting key challenges that must be overcome to enable a sustainable long-term human presence on Mars.

Some figures in this work were generated using artificial intelligence tools for conceptual illustration. These images are intended solely to support visualization of the discussed concepts and do not constitute experimental or quantitative evidence.

2. Background on the Martian regolith and building environment

The surface of Mars is covered by a fine-grained regolith formed primarily from the mechanical fragmentation and limited chemical alteration of basaltic crustal rocks. Mineralogically, Martian soil is dominated by plagioclase feldspars, pyroxenes, olivine, and iron oxides, with widespread occurrences of sulfates, sulfides, and locally clay minerals, as observed in both orbital analyses and rover-based measurements across multiple landing sites, reflecting the planet's diverse geological history [12–14]. These phases provide both mechanically competent aggregates and chemically functional components for in-situ resource utilization (ISRU).

Chemically, Martian regolith exhibits a basaltic composition, typically containing 40-50 wt% SiO_2, 10-15 wt% Al_2O_3, 10-20 wt% FeO/Fe_2O_3, 5-10 wt% MgO, and 5-8 wt% CaO, 2-5% alkali oxides (Na_2O and K_2O), ~1% TiO_2, 1-8% SO_3 and traces of Cl and salts. This composition closely resembles terrestrial basalt and underpins the interest in regolith-based geopolymers, sintered materials, and sulfur-based composites [15–17]. Accordingly, commonly used Martian regolith simulants, such as JSC Mars-1A, MGS-1, and MMS, are formulated as basaltic powders.

CheMin X-ray diffraction analyses indicate that Martian soils and fine-grained regolith contain a substantial X-ray amorphous fraction. Typical aeolian soils exhibit ~25-30 wt% amorphous material [18], whereas fine-grained sedimentary materials and altered regolith at Gale crater commonly contain ~40-60 wt% amorphous phases, depending on stratigraphy and alteration history [19]. From a construction perspective, this feature is particularly relevant, as amorphous aluminosilicates enhance reactivity in geopolymer systems, while irregular and angular particles improve packing density and interfacial bonding in sulfur-based and sintered materials. Grain sizes are predominantly silt to fine sand, shaped by long-term aeolian transport [20–22], which is favorable for extrusion-based 3D printing due to improved flowability when properly graded. Basalt-based regolith exhibits effective radiation shielding due to its relatively high density and silicate-rich composition. Radiation measurements on the Martian surface indicate exposure levels of approximately ~200–250 $mSv{\cdot}year^{-1}$, primarily due to galactic cosmic rays and solar energetic particles. For typical regolith densities (~1.5–2.0 $g\ cm^{-3}$), shielding effectiveness increases with thickness, and layers on the order of ~3 m are generally required to significantly attenuate radiation, reducing exposure toward values compatible with long-term human habitation and approaching terrestrial occupational limits (~20 $mSv{\cdot}year^{-1}$) [23–26].

Mars exhibits extreme thermal variability as a consequence of its thin atmosphere and low surface thermal inertia. Global surface temperatures range from approximately −150 °C in polar regions during winter to localized daytime maxima of +20 to +30 °C near the equator under peak insolation conditions. At low and mid-latitudes, surface temperatures display very strong diurnal cycles, with typical day-night amplitudes of 60-90 °C, locally exceeding 100 °C over dusty terrains. Pre-dawn temperatures commonly fall to −90 to −100 °C, followed by a gradual warming after sunrise and rapid cooling after sunset. These patterns have been consistently measured by Viking, InSight, Curiosity, and Perseverance and are well reproduced by diurnal temperature cycle models with root mean square errors generally below 2 °C [3].

In addition to surface variability, the lower Martian atmosphere also undergoes pronounced diurnal temperature oscillations driven by solar forcing and migrating thermal tides. Orbital observations with full local-time coverage indicate typical atmospheric diurnal variations of 10-30 °C, increasing to 40-50 °C during dusty seasons, particularly near perihelion. These temperature fluctuations are dominated by diurnal and semi-diurnal thermal tides and play a central role in regulating near-surface pressure oscillations, wind fields, and dust transport, thereby strongly coupling surface thermal forcing with atmospheric dynamics on Mars [27].

The Martian atmosphere is extremely thin; in-situ measurements from the InSight lander confirm that the present-day mean surface pressure is approximately 600-610 Pa, with superimposed diurnal and seasonal variability driven by thermal tides and CO_2 exchange with the polar caps [28,29]. Under these conditions, liquid water is thermodynamically unstable at the surface and rapidly boils, freezes, or sublimates. This challenges the use of conventional hydraulic binders and, consequently, motivates the development of water-free or water-efficient construction systems for extraterrestrial applications.

The present-day Martian atmosphere is dominated by carbon dioxide, comprising approximately 95-96 vol%, followed by molecular nitrogen (~2.6-2.7 vol%) and argon (~1.6-1.9 vol%). Minor constituents include molecular oxygen (~0.13-0.16 vol%) and carbon monoxide (~0.06-0.07 vol%), while water vapor is present only in trace amounts and exhibits strong spatial and seasonal variability [30–32]. The thin Martian atmosphere provides minimal natural radiation shielding, leading to elevated surface radiation levels. Furthermore, its low density significantly limits convective heat transfer, increasing reliance on material design and structural geometry for both thermal regulation and radiation protection.

Although storms can reach very high wind speeds in the upper atmosphere, with thermospheric horizontal winds exceeding 150-200 m s^{-1} and campaign-averaged peaks approaching ~200-240 m·s^{-1} during planet-encircling dust events, as measured in situ by the MAVEN/NGIMS instrument [33], these extreme velocities occur at altitudes above ~150 km and are decoupled from surface dynamics. Near-surface winds on Mars are generally modest under background conditions, with typical daytime speeds of 2-5 m s^{-1} and median values around 3-4 m s^{-1}. However, during regional and planet-encircling dust storms, wind speeds increase substantially, with in-situ rover observations documenting sustained winds of 15-20 m s^{-1} and gusts locally exceeding 20 m s^{-1}, sufficient to mobilize dust and sand despite the low atmospheric density [34–36]. Dust abrasion, surface erosion, and accumulation on exposed structures represent long-term durability concerns. For 3D-printed elements, surface roughness and layer interfaces may act as preferential sites for dust adhesion and erosion, emphasizing the importance of material densification and protective geometries.

Mars lacks a global magnetic field together with a very thin atmosphere, resulting in high levels of ionizing radiation at the surface. Radiation is a dominant driver in habitat design and strongly influences construction strategies. Rather than relying solely on material shielding performance, current concepts increasingly emphasize geometric and locational solutions, such as regolith overburden, partially buried structures, or subsurface construction.

Additive manufacturing (3D printing) is widely regarded as one of the most viable construction techniques for Mars due to its compatibility with robotic deployment, efficient material utilization, and high geometric flexibility. Additive manufacturing approaches, such as direct ink writing, thermoplastic extrusion, and solar- or microwave-assisted sintering, are particularly well suited for regolith-derived materials, enabling the fabrication of complex, graded, and functionally optimized structures. Moreover, the fine particle size distribution of Martian soil, together with its predominantly basaltic chemistry, supports its incorporation into printable formulations when combined with appropriate binders, including hydraulic, alkali-activated, sulfur-based systems, or binder-free sintered architectures. However, the same fine particle characteristics can also lead to high cohesiveness and poor flowability, posing challenges for feedstock handling and extrusion-based processes, such as direct ink writing, and requiring careful control of particle size distribution, additives, or processing conditions.

To mitigate radiation exposure and extreme thermal variations, two main strategies are commonly considered: surface construction with regolith shielding and subsurface or partially buried habitats. Surface structures can be 3D printed either with large wall thicknesses or with thinner walls that are subsequently covered by thick layers of regolith, thereby providing effective radiation attenuation and enhanced thermal inertia. The radiation protection requirements and habitat design strategies for Martian environments are discussed in detail in section 5. Subsurface construction, including lava tubes [37] or excavated trenches, offers inherent shielding and more stable thermal conditions, potentially reducing material durability requirements. In both cases, additive manufacturing enables on-site fabrication of liners, supports, and structural elements adapted to irregular geometries.

Although liquid water is unstable at the Martian surface, multiple lines of evidence indicate that subsurface ice is present across many regions of Mars; however, its distribution is not globally uniform, and its accessibility depends strongly on latitude, depth, and local geological conditions. Direct observations of fresh impact craters reveal clean water ice at depths of decimeters to approximately 1 m in the mid-latitudes, while global analyses combining thermal, radar, neutron, and geomorphological data indicate that such near-surface ice is regionally extensive and predictable. Physical models further show that this ice forms naturally through vapor diffusion and freezing within porous regolith, producing an ice table that is shallow at high and mid-latitudes. However, its practical accessibility for in situ resource utilization is influenced by environmental and geotechnical factors, including seasonal CO_2 frost coverage, overlying dust deposits, and variations in regolith compaction and hardness [38–40]. Hydrated minerals, such as sulfates and clays, also represent potential water sources through thermal extraction [41–43]. Other potential water acquisition pathways include atmospheric water harvesting, which is considered a secondary and supplementary route, and the exploitation of polar ice caps. However, the latter is generally regarded as highly challenging and impractical due to the extremely cold environmental conditions, the presence of seasonal CO_2 ice cover, and significant operational and logistical constraints [44].

Table 1 summarizes the main physical, chemical, and mechanical properties of Martian regolith relevant to construction and additive manufacturing applications. The values reported highlight the strong variability of Martian regolith properties, which depend on location, particle size distribution, and degree of alteration. These characteristics have direct implications for additive manufacturing processes, influencing material reactivity, flow behavior, densification, and long-term structural performance.

Perchlorate salts can also be present in Martian regolith, typically at concentrations on the order of 0.5-1.0 wt%. Although relatively minor in

Table 1
Physical, chemical, and mechanical properties of Martian regolith relevant to construction.

Property	Typical range/ value	Notes	Reference
SiO_2 content	40-50 wt%	Basaltic composition	[8–10]
Al_2O_3 content	10-15 wt%	Supports geopolymerization	[8–10]
FeO/Fe_2O_3	10-20 wt%	Influences microwave absorption	[8–10]
MgO	5-10 wt%	Basaltic mineralogy	[8–10]
CaO	5-8 wt%	Relevant for binding phases	[8–10]
Alkali oxides (Na_2O + K_2O)	2-5 wt%	Important for activation reactions	[8–10]
SO_3	1-8 wt%	Indicates sulfur availability	[8–10]
Amorphous content	~25-60 wt%	Enhances reactivity	[11,12]
Grain size	Silt to fine sand	Favorable for extrusion	[13–15]
Particle shape	Angular, irregular	Improves interlocking	[13–15]
Bulk density	~1100-1600 kg/m^3	Depends on compaction	[16–19]
Solid density	~2900-3100 kg/m^3	Typical basaltic values	[16–19]
Porosity	~30-50% (loose)	Affects strength/ permeability	[13–15]
Cohesion	~0.2-5 kPa	Weakly cohesive soil	[27–29]
Friction angle	~30-40°	Similar to sand	[27–29]
Thermal conductivity	~0.05-0.2 W/ m·K	Strongly density-dependent	[4]
Specific heat capacity	~600-900 J/ kg·K	Typical silicate range	[4]
Surface temperature range	−150 to +30 °C	Extreme variability	[4]
Atmospheric pressure	~600 Pa	Affects processing	[21,22]

proportion, their presence introduces important chemical considerations for in-situ construction materials. These compounds are strong oxidizing agents and exhibit hygroscopic behavior, which may influence both the durability and processing of binder systems.

In the case of alkali-activated materials (discussed in Section 4.1), perchlorates may affect reaction kinetics and microstructure development. Due to their hygroscopic nature, these salts can alter local moisture availability, a critical parameter for geopolymerization reactions. This may lead to variations in setting time, degree of reaction, and resulting porosity. Additionally, the incorporation of soluble salts into the pore structure may impact long-term durability through crystallization-induced stresses and modifications in ionic transport properties.

For sulfur-based materials (discussed in Section 4.2), which rely on thermoplastic behavior rather than hydration reactions, direct chemical interaction with perchlorates is expected to be limited under ambient conditions. However, the oxidative nature of perchlorates - particularly under elevated temperatures or in the presence of catalytic mineral phases - may contribute to long-term degradation mechanisms such as sulfur oxidation or microstructural embrittlement. These effects may be further exacerbated by thermal cycling and radiation, although experimental data under representative Martian conditions remain scarce.

Overall, while current understanding is largely based on terrestrial analogs, the interaction between perchlorates and binder systems represents a key uncertainty for Martian construction. Further experimental studies under relevant environmental conditions are required to quantify their impact on material performance and long-term structural reliability.

Lastly, the mineralogical composition of Martian regolith varies significantly across different landing sites, reflecting diverse geological histories that include volcanic, sedimentary, and aqueous processes. These variations may directly affect the suitability of regolith as feedstock for additive manufacturing, influencing sintering behavior, reactivity in binder-based systems, and the resulting mechanical performance of fabricated structures.

3. Review methodology and literature selection

A systematic literature search was conducted using the Web of Science database. The search strategy was designed to capture studies at the intersection of in situ resource utilization (ISRU), Martian construction, and additive manufacturing technologies. The primary search string used was: ("in situ resource utilization" OR "ISRU") AND ("Mars" OR "Martian") AND ("additive manufacturing" OR "3D printing"), which resulted in a total of 56 publications.

This dataset was used primarily to support the development of Section 4, which focuses on additive manufacturing technologies and material systems for Martian construction. The initial set of articles was screened based on titles and abstracts to exclude studies not directly related to construction materials, additive manufacturing processes, or infrastructure development for Mars applications. Additional exclusion criteria included works focused solely on life-support systems, propulsion, or unrelated space technologies. The remaining studies were selected for full-text analysis.

The selected publications were categorized according to (i) material systems (e.g., geopolymer-based, sulfur-based, and sintered regolith), (ii) additive manufacturing techniques (e.g., extrusion-based processes, thermoplastic deposition, and sintering approaches), and (iii) application scale (e.g., structural elements, shielding systems, or small-scale components). To ensure a consistent comparison across the literature, a common evaluation framework was adopted based on key engineering metrics, including mechanical performance (compressive and flexural strength), process requirements (temperature, pressure, and energy demand), ISRU compatibility, construction scalability, and durability under Martian environmental conditions.

In addition to this structured search, complementary targeted searches were conducted to support other sections of the paper, particularly those addressing Martian soil properties and environmental conditions. These additional searches expanded the overall body of literature considered in this review, contributing to a more comprehensive understanding of the topic.

4. 3D printing technologies and materials

Additive manufacturing represents one of several competing paradigms for extraterrestrial construction. Compared to prefabricated systems, 3D printing offers significant reductions in launch mass through in-situ resource utilization, but at the cost of increased system complexity and energy demand. Relative to modular assembly approaches, it enables greater geometric flexibility and structural integration, though often with lower process reliability and longer construction times. These trade-offs highlight that 3D printing should not be considered a universally optimal solution, but rather one that becomes increasingly advantageous as infrastructure maturity and ISRU capabilities improve.

It is important to note that the construction approaches discussed in this section primarily address the fabrication of structural and shielding elements. The realization of fully functional habitats requires the integration of externally supplied systems, including pressure vessels, airlocks, and internal infrastructure, which remain beyond the scope of current regolith-based additive manufacturing capabilities.

The feasibility of large-scale additive manufacturing on Mars is intrinsically linked to the availability of water and binders, the nature of regolith-based material systems, and the technological maturity and scalability of the printing process. This section presents a critical comparison of the principal techniques and materials proposed for autonomous, especially for large-scale construction under Martian environmental constraints.

4.1. Extrusion-based additive manufacturing: printing at low temperatures

Extrusion-based techniques, including direct ink writing and contour crafting, represent the most experimentally mature additive manufacturing approaches for extraterrestrial construction, having been extensively investigated at the laboratory scale and demonstrating clear potential for scalability [45–51]. These methods rely on the deposition of a regolith-based paste or mortar, typically formulated with geopolymeric binders.

While these approaches enable the fabrication of large, monolithic structures with relatively high build rates, they typically rely on water or chemically reactive binders, thereby introducing significant logistical complexity. Water scarcity, particularly during the initial exploration stage, rapid evaporation under low atmospheric pressure, the requirement for alkaline activators that are difficult to produce in situ and would likely need to be transported from Earth, and challenges associated with curing under reduced gravity and extremely low temperatures remain critical constraints.

Ma et al. [45] aimed to demonstrate the feasibility of producing structural building components directly from Martian regolith simulants using 3D printing. The primary objective of their work was not only to validate the printability of regolith-based materials, but also to overcome the intrinsic brittleness of geopolymer systems through architectural design, thereby enabling damage-tolerant structures suitable for extraterrestrial environments. To this end, the authors employed a basaltic Martian regolith simulant (HIT-MRS-1) engineered to reproduce the mineralogical and chemical characteristics of Martian soil. The simulant is rich in SiO_2 (~48 wt%), Al_2O_3 (~14 wt%), Fe_2O_3 (~17 wt %), MgO (~6-7 wt%), and CaO (~8 wt%), and contains a significant amorphous fraction (~28 wt%), which is critical for alkali activation. The regolith simulant was activated using a sodium silicate and NaOH alkaline solution to form a geopolymer binder and reinforced with 3 vol

% short basalt fibers. While monolithic geopolymer specimens exhibited limited strain capacity, the introduction of bioinspired architectures, including suture, helical, honeycomb, and tubular patterns, dramatically enhanced damage tolerance. In particular, helical structures achieved compressive strengths of approximately 32 MPa, whereas suture-inspired geometries, despite lower compressive strengths (~9 MPa), exhibited exceptionally high ultimate strains (~11-14%), highlighting the dominant role of structural architecture over intrinsic material brittleness. From an ISRU perspective, although basalt fibers could, in principle, be produced on Mars through melting and fiberization of basaltic regolith, the overall system remains strongly dependent on Earth-derived chemical inputs, namely alkaline activators (NaOH and sodium silicate), whose large-scale production on Mars is currently highly challenging. In addition, geopolymer systems require controlled curing over hours or days and are highly sensitive to pressure and temperature conditions.

Hedayati and Stulova [46] investigated the feasibility of 3D printing Martian habitats using regolith-based geopolymer mixtures under simulated Martian environmental conditions, with particular emphasis on the effects of low temperature and low atmospheric pressure on curing behavior and mechanical performance. A basaltic Martian regolith simulant was combined with a water-sodium silicate (Na_2SiO_3) binder and processed using both molding and extrusion-based additive manufacturing. The results clearly demonstrated that Martian surface conditions are highly detrimental to geopolymerization, as the combined effects of low temperature (~2 °C) and low pressure (10^4-10^3 Pa) caused rapid water evaporation and premature freezing, preventing proper gel formation and leading to highly porous, mechanically weak materials. Satisfactory mechanical performance was achieved only under Earth-like or tightly controlled curing conditions, particularly at elevated temperature (~60 °C) and atmospheric pressure, where flexural strengths of up to ~9 MPa were obtained; in all cases, 3D-printed specimens exhibited approximately 20% lower strength than molded counterparts. These findings provide strong experimental evidence that water-based alkali-activated systems are incompatible with direct, unprotected surface construction on Mars. Furthermore, the proposed binder system relies heavily on water and sodium silicate, both of which are scarce and energetically costly to obtain or synthesize on Mars, rendering large-scale application questionable from an in-situ resource utilization (ISRU) perspective.

Under Martian surface conditions, these requirements would necessitate the use of pressurized and thermally controlled environments, which are expected to become feasible only after the establishment of a basic infrastructure. In practice, 3D printing operations would need to be carried out inside a pressurized inflatable chamber equipped with controlled thermal management. Such a configuration is conceptually illustrated in Fig. 1, inspired by the design proposed by AI SpaceFactory as part of NASA's 3D Printed Habitat Challenge. In this concept, the design known as MARSHA (Mars Habitat) consists of cylindrical modules intended as habitable structures for human use on the Martian surface. The construction approach relies exclusively on in-situ resources, combining basalt fiber - derived from Martian surface materials - with a recyclable biopolymer produced from plant-based feedstocks cultivated on Mars [52].

Such structures must be maintained throughout the entire curing stage, typically for at least 24-72 h, in order to achieve approximately 60-80% of the final mechanical strength, as commonly obtained at room temperature under terrestrial conditions. However, under Martian conditions, the curing process is expected to be significantly slower due to the combined effects of low ambient temperatures, reduced atmospheric pressure, limited water availability, and enhanced evaporative losses. These factors hinder geopolymerization by reducing dissolution kinetics, limiting ion transport, and promoting premature drying, which may delay strength development and compromise microstructural integrity if not properly controlled.

To mitigate these effects, curing efficiency may be significantly enhanced by heating either the inflatable chamber as a whole or, more efficiently, the printing head and the freshly deposited layers through localized thermal systems (e.g., electrical resistive heaters or microwave-based heating) [53], operating within a temperature range of 40-80 °C [54–56]. This approach promotes improved interlayer adhesion, reduces the formation of cold joints, and can shorten the curing period to only a few hours. After completion of the curing process, the chamber should be gradually depressurized in order to prevent rapid water evaporation and associated microstructural damage, such as excessive shrinkage and microcracking within the geopolymeric matrix.

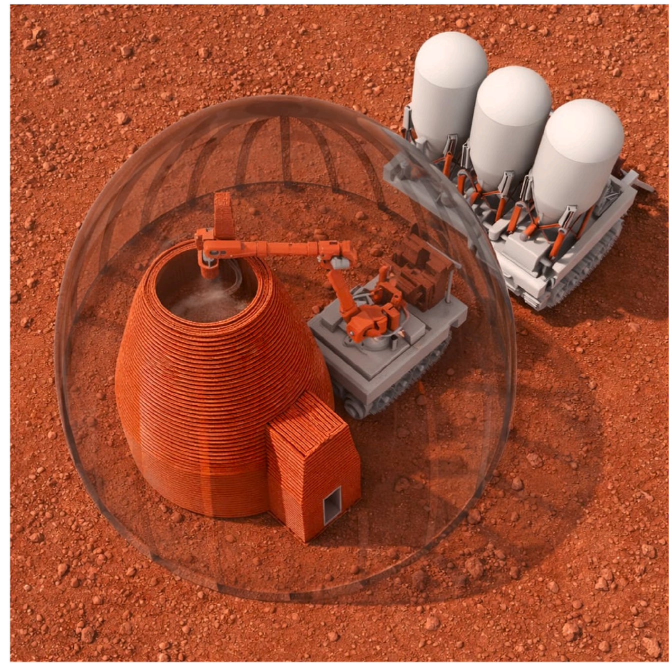

Fig. 1. Schematic representation of 3D printing of a regolith-based mixture using a direct ink writing (DIW) technique under controlled temperature and pressure conditions. The configuration is inspired by the MARSHA habitat concept proposed by AI SpaceFactory.

The low gravitational acceleration on Mars (≈3.71 m s^{-2}) may benefit the construction of habitats using alkali-activated regolith, as it significantly reduces self-weight-induced stresses during and after the printing process. This condition is expected to enhance the geometric stability of freshly deposited layers, lower the risk of plastic collapse, and enable the fabrication of taller walls or larger spans prior to full hardening. In addition, reduced gravity may limit particle segregation and help preserve the early-age cohesion of the extruded material. Conversely, the same environment introduces challenges related to layer compaction, interlayer contact, and early strength development, requiring careful control of rheology and curing conditions. Table 2 summarizes the principal advantages and limitations of extrusion-based additive manufacturing using alkali-activated binders.

Although this approach is technologically mature and capable of producing large-scale structures, its dependence on water, chemical activators, and controlled curing conditions represents a major limitation under direct Martian surface environments.

The Mars Dune Alpha habitat (see Fig. 2), developed by ICON for NASA's CHAPEA program, exemplifies large-scale extrusion-based construction using a proprietary cementitious material (named Lavacrete), a printable concrete-like composite specifically engineered for terrestrial additive manufacturing. Although this analog system does not directly employ regolith, it serves as a critical testbed for validating large-scale printing processes, structural performance, and operational workflows relevant to future in-situ resource utilization applications. In this context, NASA research indicates that extraterrestrial construction

Table 2
Summary of the main advantages and limitations of extrusion-based additive manufacturing using alkali-activated (geopolymer) binders for Martian construction, considering ISRU compatibility, processing conditions, mechanical performance, and environmental constraints.

Aspect	Advantages	Disadvantages
ISRU compatibility	Uses regolith as main aggregate	Requires alkaline activators (NaOH, Na_2SiO_3) likely imported and high volumes of water
Process maturity	Most experimentally developed and scalable	Strong dependence on controlled conditions
Mechanical performance	Good compressive strength; tunable via design	Brittle behavior without architectural optimization
Environmental compatibility	Works at low temperatures	Water evaporation/freezing under Mars conditions
Construction logistics	Suitable for large monolithic structures	Requires pressurized and heated enclosure
Curing	Established knowledge from Earth systems	Curing time remains a constraint (hours-days), albeit mitigated by thermal acceleration

Fig. 2. Large-scale extrusion-based additive manufacturing of a cementitious material using ICON's robotic printing system (Vulcan) for the construction of the Mars Dune Alpha habitat prototype. The structure is fabricated layer-by-layer under terrestrial conditions as part of NASA's CHAPEA program, serving as an analog for future regolith-based construction on Mars (Image credit: ICON/NASA).

materials will likely rely on regolith as the primary aggregate, combined with locally sourced binders such as water or other extractable compounds, enabling the production of concrete-like materials or ceramic structures through thermal or laser-based processing [57]. From a materials perspective, this highlights a key transition from Earth-based cementitious systems toward regolith-driven composites, while maintaining similar extrusion-based fabrication principles. However, as discussed throughout this section, such approaches remain fundamentally constrained by binder availability, curing requirements, and environmental compatibility under Martian surface conditions.

4.2. Thermoplastic extrusion: printing at middle temperatures

The use of thermoplastic extrusion on Mars enables the usage of another class of construction material that eliminates the need for water, commonly referred to as sulfur concrete or Martian concrete. The term Martian Concrete refers specifically to sulfur-based construction materials composed of molten sulfur acting as a thermoplastic binder combined with Martian regolith simulant as aggregate, as originally proposed in previous studies (e.g., Wan et al. [47]). More generally, such materials may also be described as sulfur-regolith composites, a broader term that encompasses any composite system formed by combining sulfur with regolith-derived particles, regardless of processing route or formulation. In this manuscript, Martian Concrete is used to denote optimized, concrete-like formulations with structural performance comparable to terrestrial construction materials, while sulfur-regolith composite is used as a generic descriptor.

Similar to regolith, sulfur is also abundant on Mars, where it occurs mainly as sulfates and sulfides widely distributed in basaltic rocks and sedimentary deposits. This availability makes sulfur particularly attractive for ISRU. However, sulfur-based concrete exhibits several limitations. Under certain conditions, particularly in environments characterized by pronounced temperature fluctuations, sulfur-based matrices tend to become increasingly brittle and to experience progressive strength degradation over time, which constrains their suitability for long-term structural applications [58].

Sulfur concrete fundamentally differs from hydraulic binders in that hardening occurs by thermal solidification of molten sulfur, rather than hydration reactions. When heated above its melting point (typically in the 120-140 °C range) and subsequently quenched, sulfur solidifies into a continuous thermoplastic matrix that binds the aggregate particles. Strength development is therefore extremely rapid and controlled primarily by thermal history, sulfur crystallization kinetics, and interfacial bonding [59,60].

Producing structures based on sulfur concrete on Mars would require a production process markedly different from that of terrestrial Portland cement concrete, as well as strategic choices to overcome the intrinsic limitations of the material. The process would be essentially thermoplastic in nature, bearing greater similarity to polymer or asphalt-based industries than to conventional hydraulic concrete technologies.

Sulfur concrete has been extensively investigated for terrestrial applications [61,62]. While some studies have explored its potential under extraterrestrial conditions, most of these works rely on lunar regolith [63,64], with only scarce studies employing Martian regolith simulants.

Shahsavari et al. [65] experimentally investigated sulfur-based construction materials formulated with Martian regolith-representative compositions to assess their suitability for extraterrestrial infrastructure. Strength development occurred rapidly upon cooling, confirming that hardening was governed by thermal solidification rather than hydration reactions. The resulting sulfur-regolith composites achieved compressive strengths on the order of 20-40 MPa, depending on sulfur content and oxide composition. Mechanical performance was strongly composition-dependent, with iron-rich systems generally exhibiting enhanced strength and reduced porosity. Microstructural analyses revealed limited chemical interactions between sulfur and regolith oxides, leading to the formation of iron- and aluminum-sulfate phases primarily at the sulfur-aggregate interface. When confined to the interfacial region, these reaction products were found to be beneficial, improving interfacial bonding and locally densifying the microstructure by partially filling microvoids. However, excessive sulfate formation or poor thermal control could adversely affect pore structure through localized microcracking, indicating the need for careful optimization of composition and processing conditions. Sublimation experiments conducted under Mars-relevant temperatures further demonstrated that sulfur mass loss is negligible at typical Martian surface conditions, supporting the long-term stability of sulfur-regolith composites for Mars construction applications.

In the study of Ilerioluwa Giwa et al. [48], elemental sulfur was combined with a high-fidelity Martian regolith simulant (MGS-1), characterized by a basaltic composition rich in SiO_2 (~43 wt%), Al_2O_3 (~13 wt%), FeO (~11 wt%), MgO (~15 wt%), and CaO (~7 wt%). The mixture contained approximately 33 wt% sulfur, with the remainder consisting of regolith particles. A modified formulation incorporating dicyclopentadiene (DCPD) at about 10 wt% relative to the sulfur content was also investigated. Mixing and extrusion were conducted at 130-150 °C, while the regolith was preheated to ~170 °C to prevent premature solidification; hardening occurred through rapid cooling after extrusion.

In terms of absolute mechanical performance, mold-cast specimens reached compressive strengths on the order of 35-40 MPa within 7 days, while DCPD-modified mixtures achieved compressive strengths of approximately 58-60 MPa. For additively manufactured elements, the flexural strength of unmodified mixtures was about 9-10 MPa, increasing to ~13 MPa with DCPD modification. Remarkably, 75-85% of these ultimate flexural strengths were attained within the first 12 h, underscoring the exceptional early-age performance of sulfur-based binders. Under vacuum at ambient temperature, the mixtures maintained both compressive and flexural performance. At elevated temperature (55 °C) under vacuum, sulfur sublimation caused mass loss and moderate strength reduction in unmodified SRC; this effect was significantly mitigated by DCPD, which improved strength retention and microstructural stability.

Wan et al. [47] developed and systematically evaluated a sulfur-based "Martian Concrete" composed of molten sulfur and a Martian regolith simulant (JSC Mars-1A), aiming to demonstrate a viable water-free construction material for Mars. Through an extensive experimental campaign, sulfur contents between 35 and 60 wt% were investigated, identifying an optimal mixture of approximately 50% sulfur and 50% regolith, which achieved compressive strengths exceeding 50 MPa within 24 h of cooling. Mechanical characterization included unconfined compression, three-point bending, splitting tensile, and fracture energy tests, revealing strength levels significantly higher than those of sulfur concrete made with conventional sand. Microstructural analyses showed that the fine particle size distribution and the metal-rich composition of the Martian regolith simulant promote limited chemical interactions with sulfur, leading to the formation of sulfates and polysulfates that reduce porosity and enhance bonding. The study further demonstrated that material performance can be improved by recasting and by applying pressure during casting, owing to sulfur shrinkage upon cooling. In parallel, the mechanical response of Martian Concrete was successfully simulated using the Lattice Discrete Particle Model (LDPM), showing excellent agreement with experimental results under compression, bending, and fracture. Overall, the work establishes sulfur-based Martian Concrete as a high-strength, rapidly curing, recyclable, and ISRU-compatible construction material well suited to the dry and cold Martian environment.

Wang and Snoeck [60] provide a comprehensive review of sulfur concrete with particular emphasis on its relevance for sustainable and extraterrestrial construction. The authors highlight that sulfur concrete fundamentally differs from hydraulic systems, as hardening occurs through physical solidification of molten sulfur rather than hydration, enabling rapid strength development, water-free processing, and full recyclability through remelting. The review summarizes experimental studies demonstrating that sulfur concrete produced with basaltic aggregates and planetary regolith simulants can achieve mechanical properties comparable to conventional concrete, while offering enhanced resistance to acids, salts, and aggressive environments. Remaining challenges identified include sensitivity to thermal cycling, sulfur phase transformations, long-term durability, and the need for optimized thermal management during fabrication and service.

The first stage in the 3D printing of habitats using sulfur concrete involves the acquisition and preparation of raw materials. Sulfur could be extracted from sulfate-bearing deposits present in the Martian regolith (e.g., calcium, magnesium, or iron sulfates) through thermal reduction or relatively simple chemical processes. In parallel, local regolith would be excavated, sieved, and granulometrically classified, prioritizing particle size fractions that maximize packing density and mechanical performance. Unlike Portland cement systems, the aggregate in sulfur concrete cannot be considered inert, as its mineralogical composition directly affects interfacial bonding with molten sulfur.

The subsequent stage would be the production of the sulfur binder, which requires heating sulfur above approximately 120-140 °C to ensure adequate fluidity. This step is particularly critical on Mars because it would occur under low-pressure conditions without appropriate control. While sulfur loss by volatilization was considered negligible under controlled laboratory conditions by Shahsavari et al. [65], Martian surface conditions - characterized by reduced pressure and significant thermal gradients - may enhance volatilization effects and contribute to localized defects during large-scale processing if not properly controlled. Once molten, sulfur would be rapidly mixed with preheated regolith aggregates, forming a viscous composite that must be shaped before solidification.

In thermoplastic extrusion-based 3D printing, molten sulfur concrete can be deposited layer by layer in a manner analogous to asphalt paving or polymer additive manufacturing. Heating may be provided by resistive heaters or microwave systems integrated within the printing nozzle; however, an internal mixing mechanism is also required to ensure uniform heat distribution and proper homogenization of the molten material prior to extrusion [53]. The material would solidify rapidly after deposition, eliminating the need for curing. However, this strategy requires precise control of nozzle temperature and careful thermal management to mitigate cracking during cooling. A conceptual illustration of this approach is shown in Fig. 3. Side and bottom trowels can be integrated into the nozzle head to enable the fabrication of curved geometries, such as the closing layers of dome-shaped structures shown in the same figure. The temperature of trowels could also be regulated for better quenching control. Within this context, dome geometries have been widely explored as efficient and structurally optimized solutions for both terrestrial and Martian habitats, as demonstrated in projects by WASP [66] and Northwestern University in collaboration with Skidmore, Owings & Merrill (SOM) [67,68]. This preference is primarily attributed to the favorable stress distribution inherent to dome structures, which minimizes tensile stresses by promoting compression-dominated load transfer, as well as their enhanced resistance to internal pressurization. These characteristics make dome-based configurations particularly suitable for habitats subjected to significant pressure differentials in extraterrestrial environments.

An alternative and potentially more robust construction approach involves the production of prefabricated blocks or modular elements. In this strategy, molten sulfur-based concrete can be cast into metallic or ceramic molds, solidifying within minutes to form structural units. These elements can then be robotically assembled to construct walls, vaults, or radiation-shielding structures. This approach offers simplicity, rapid processing, and mechanical robustness, making it particularly suitable for early-stage infrastructure such as protective barriers, foundations, and shielding systems. A conceptual illustration of this construction strategy is presented in Fig. 4.

Sun et al. [69] investigated sulfur-bonded materials exclusively in the form of cast building blocks, without employing additive manufacturing or extrusion techniques, with the specific aim of

Fig. 3. Conceptual illustration of an autonomous 3D printing system constructing a Martian shelter using sulfur-based concrete, deposited layer by layer from locally sourced materials. The design is inspired by previously proposed dome-based habitat concepts developed for terrestrial and extraterrestrial applications [66,67].

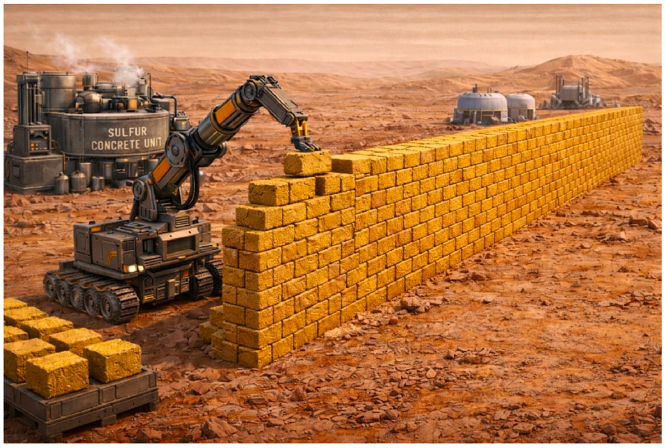

Fig. 4. Conceptual illustration of an autonomous robotic system constructing a protective wall from sulfur concrete blocks on Mars.

developing construction solutions for Martian habitats. In their study, elemental sulfur was used as a thermoplastic binder and combined with a Martian soil simulant and selected oxide aggregates to fabricate sulfur-based bricks hardened solely through thermal solidification. The Martian simulant exhibits a basaltic composition representative of Martian regolith, consisting primarily of SiO_2 (~45 wt%), Al_2O_3 (~10 wt%), Fe_2O_3 (~18 wt%), CaO (~6 wt%), and MgO (~8 wt%). Sulfur-simulant mixtures were produced with simulant contents ranging from 0 to 20 wt%, while additional formulations incorporated individual oxides (SiO_2, Al_2O_3, and Fe_2O_3) as aggregates. The results showed that increasing simulant content enhanced mechanical performance, with an optimum at 20 wt% MS, yielding compressive strengths of approximately 25 MPa, whereas oxide-based systems achieved even higher strengths, reaching ~33 MPa with SiO_2 and up to ~39 MPa with Fe_2O_3. The study further demonstrated that sulfur can effectively function as both a structural binder and a bonding agent between blocks, achieving joint strengths comparable to those of terrestrial masonry mortars, and highlighted the recyclability of sulfur blocks through repeated melting and casting cycles. Despite convincingly demonstrating the feasibility of sulfur-based modular construction elements, the study did not evaluate durability under vacuum or Mars-relevant environmental conditions, leaving long-term performance under extraterrestrial exposure as an important open research gap.

From a critical perspective, although the combined abundance of regolith and sulfur on Mars supports sulfur concrete as an ISRU-compatible, water-free construction material, important limitations arise during both processing and long-term service. Beyond production, durability concerns dominate the long-term performance of sulfur concrete. In service, sulfur-based matrices exhibit pronounced thermal sensitivity, including increased brittleness at low temperatures and susceptibility to damage under repeated thermal cycling, both of which are exacerbated by Martian diurnal and seasonal temperature fluctuations. Łaźniewska-Piekarczyk et al. [59] tested sulfur concrete for freeze-thaw resistance and found that after 150 freeze-thaw cycles the compressive strength dropped by >30 %, far more than allowable limits in conventional standards, indicating poor durability under cyclic thermal stress. The material also exhibited sudden failure and shattering modes typical of brittle fracture in compression tests. Additionally, the frequent reliance on Earth-supplied chemical modifiers to mitigate these drawbacks reduces the overall autonomy of sulfur concrete as an ISRU solution. Consequently, while sulfur concrete remains a promising binder for early-stage Martian construction, future research should focus on controlling sulfur loss during processing, improving resistance to low-temperature brittleness and thermal fatigue, and developing modifier-free formulations compatible with realistic Martian environmental conditions.

Table 3 summarizes the principal advantages and limitations of sulfur-based thermoplastic extrusion for Martian construction. While this approach offers a water-free, rapidly solidifying, and potentially ISRU-compatible solution, its implementation is constrained by thermal processing requirements, sensitivity to temperature fluctuations, and long-term durability concerns under cyclic Martian environmental conditions.

4.3. Solar sintering-based additive manufacturing: printing at high temperatures

Solar-based additive manufacturing has gained increasing attention as a viable pathway for in-situ resource utilization (ISRU) on Mars. In particular, solar-driven sintering exploits the intrinsic mineralogy of Martian regolith, enabling structural consolidation through partial melting and viscous flow of native silicate phases. In this context, Deng et al. [70] demonstrated that basaltic regolith simulants rich in plagioclase and pyroxene readily undergo liquid-phase-assisted sintering, achieving high densification and enhanced mechanical strength without the use of additives, provided that particle size distribution and thermal conditions are carefully controlled.

The feasibility of solar sintering on Mars must be assessed in light of the Martian radiation environment. Due to its greater heliocentric distance, Mars receives a lower solar irradiance than Earth, with an average solar flux at the top of the Martian atmosphere of approximately 590 W m^{-2} [71,72], compared to about 1361 W m^{-2} at Earth orbit [73,74]. Surface irradiance on Mars is highly dynamic, ranging from a daily mean of approximately 308 W/m^2 during average conditions to as low as 95 W/m^2 during periods of peak atmospheric opacity [71]. On Earth, surface solar irradiance can approach ~1000 W m^{-2} under clear-sky conditions at local solar noon [75]; however, the global mean surface irradiance is much lower (~160-170 W m^{-2}). This reduction results from geometric averaging over the spherical Earth (a factor of 1/4), planetary albedo (~30%), atmospheric absorption and scattering, cloud cover, and the angular dependence of solar incidence [76].

However, unlike the Earth's environment, the extremely thin Martian atmosphere results in negligible convective heat losses, such that radiative heating dominates thermal exchange [77]. This fundamental difference means that, despite the lower incident solar flux, the thermal efficiency of concentrated solar sintering on Mars can be comparable to terrestrial conditions, where convective and conductive losses are substantial.

The integration of solar sintering with additive manufacturing naturally follows from these considerations. Conceptually, the process can be viewed as a radiative analogue of laser powder bed fusion, in which concentrated sunlight replaces the laser as a movable heat source. One particularly promising configuration involves a robotic 3D printing system that deposits dry regolith powder layer by layer, followed

Table 3
Summary of the main advantages and limitations of sulfur-based thermoplastic extrusion (sulfur concrete) for Martian construction, considering ISRU compatibility, processing requirements, mechanical performance, and durability under Martian environmental conditions.

Aspect	Advantages	Disadvantages
ISRU compatibility	Fully water-free; sulfur available on Mars	Sulfur extraction still requires processing
Mechanical performance	High early strength; fast solidification	Brittle behavior, especially at low temperatures
Processing	No curing required; rapid construction	Requires continuous heating (120-150 °C)
Durability	Good chemical resistance	Poor thermal cycling resistance
Recyclability	Fully recyclable by remelting	Potential sulfur sublimation at high temperatures
Construction approach	Compatible with extrusion and prefabrication	Thermal cracking risk during cooling

immediately by localized solar sintering. In such a system, a thin layer of sieved regolith is mechanically spread or dispensed, and a focused solar beam (which can be generated using a Fresnel lens) is scanned across the surface to selectively sinter the deposited layer. The build platform is then lowered, and the process is repeated, enabling the fabrication of three-dimensional objects without binders or imported feedstocks. Concerning the construction speed, Karacasulu et al. [78] showed the possibility of rapid heating of Martian regolith simulants via fast firing, promoting densification over crystallization by delaying devitrification of basaltic glass phases, thereby maintaining a lower melt viscosity and enhancing viscous flow during consolidation. These thermal conditions closely resemble those expected during concentrated solar sintering, where heating rates are inherently high and exposure times are short. Consequently, solar-based layer-by-layer sintering is likely to benefit from the same kinetic advantages observed in fast-firing experiments.

Particle size distribution further governs the effectiveness of this integrated printing approach. Fine regolith particles improve packing density and reduce diffusion distances, lowering the energy threshold required for liquid-phase formation. As demonstrated by Deng et al. [70], reducing the median particle size by approximately one order of magnitude can more than double densification efficiency and significantly enhance mechanical strength. In a solar-based 3D printing context, such improvements directly translate into reduced sintering time per layer and, consequently, lower demands on concentrator size and tracking precision. Fig. 5 shows a conceptual view of Martian habitat shelters consolidated via solar sintering. The design was inspired by a conceptual Mars habitat developed by the architecture firm HASSELL in collaboration with the structural engineering consultancy Eckersley O'Callaghan (EOC), as part of NASA's 3D Printed Habitat Challenge [79]. In this concept, an external shell constructed from locally sourced Martian regolith is envisioned to be autonomously fabricated prior to crew arrival, enclosing a series of inflatable interior modules designed to support human habitation and operations on the Martian surface.

From a system-level perspective, the combination of dry regolith deposition and simultaneous solar sintering offers several advantages for Martian construction. The layer-by-layer approach allows precise control of geometry and internal porosity. By modulating energy input and layer thickness, it becomes possible to fabricate functionally graded structures, with dense, load-bearing regions and more porous layers optimized for thermal insulation and radiation shielding. Table 4 summarizes the principal advantages and limitations of solar sintering-based additive manufacturing. This approach enables fully binder-free construction using in-situ materials and renewable solar energy; however, its effectiveness depends on solar irradiance, process control, and the ability to achieve consistent densification under variable environmental conditions.

Fig. 5. Conceptual illustration of ISRU-based Martian habitat structures fabricated through layer-by-layer regolith deposition using additive manufacturing, followed by solar-driven sintering for structural consolidation. The design is inspired by conceptual Mars habitat proposals developed by HASSELL in collaboration with Eckersley O'Callaghan (EOC) within NASA's 3D Printed Habitat Challenge.

Table 4
Summary of the main advantages and limitations of solar sintering-based additive manufacturing for Martian construction, considering ISRU compatibility, energy requirements, process control, and scalability under Martian environmental conditions.

Aspect	Advantages	Disadvantages
ISRU compatibility	Fully binder-free	Requires high solar concentration systems
Energy source	Uses renewable solar energy	Dependent on dust and irradiance variability
Material performance	Dense, strong ceramic-like structures	Limited control of microstructure
Scalability	Suitable for large structures	Requires precise solar tracking
Process complexity	No material transport needed	Slow layer-by-layer process

In summary, the performance of solar sintering systems demonstrated under terrestrial conditions may not be directly transferable to Mars. Environmental factors such as dust accumulation on optical components, fluctuations in solar irradiance, uneven regolith surfaces, and reduced gravity can affect energy concentration, thermal gradients, and melt pool stability. These effects may lead to reduced process efficiency, variability in material properties, and lower reliability compared to controlled Earth-based experiments, highlighting the need for robust system design and adaptive process control for Martian applications.

4.4. *Microwave sintering-based additive manufacturing: printing at high temperatures*

Microwave sintering represents another binder-free and water-free approach, relying on the intrinsic microwave absorptivity of iron-rich Martian regolith to induce localized melting and particle fusion. From an ISRU standpoint, this technique is also particularly attractive, as it minimizes dependence on imported materials. However, the process is energy-intensive, and controlling thermal gradients is challenging, often leading to microcracking and brittle behavior. Build rates are generally lower than extrusion-based methods, and scaling microwave systems to habitat-sized structures remains a major engineering challenge. Despite these limitations, microwave sintering is considered a strategically important medium-term solution, especially for radiation shielding elements and infrastructure components where mass and durability are prioritized over architectural complexity. Fig. 6 illustrates a conceptual view of a building structure fabricated using a microwave-based 3D printing system.

While the literature is relatively abundant regarding the use of microwave heating for processing lunar regolith [80–83], studies addressing Martian regolith remain comparatively scarce. Lunar regolith is predominantly basaltic and anorthositic, containing iron-bearing phases such as ilmenite ($FeTiO_3$), ferrous silicates, and nanophase metallic iron formed through space weathering. These phases exhibit measurable dielectric losses under microwave irradiation, which has enabled extensive experimental investigation of microwave-induced heating and sintering in lunar soil simulants.

In contrast, Martian regolith exhibits a more compositionally diverse assemblage, including iron oxides, amorphous silicate phases, sulfates, chlorides, and locally phyllosilicates, reflecting a complex geological and geochemical history. As a result, its interaction with microwave radiation is expected to be strongly dependent on local mineralogy, particularly the iron content and speciation (Fe^{2+}/Fe^{3+}), and the relative proportion of amorphous phases [84–86], which exhibit

Fig. 6. Conceptual illustration of a building structure fabricated using a microwave-based 3D printing system fed with regolith-derived materials, which are heated through volumetric microwave absorption to form a tunnel-like structure.

significantly higher dielectric and magnetic losses, leading to enhanced microwave absorption.

Consequently, microwave heating of Martian regolith is expected to produce heterogeneous thermal responses that may vary significantly among landing sites. Compositionally homogeneous batches are required to ensure reproducible and uniform heating behavior. Therefore, this variability highlights the need for systematic experimental studies to evaluate microwave-regolith interactions under controlled conditions and to assess their implications for sintering, consolidation, and additive manufacturing processes in Martian ISRU applications.

Conceptual habitat designs developed by HASSELL in collaboration with Exploration Architecture Corporation (EOC) reference the use of microwave-based processing as a potential ISRU technique for regolith consolidation and construction on Mars. In these concepts, microwave heating is proposed as a volumetric energy delivery method for in situ material processing, although the approach is presented at a conceptual level and is not supported by experimental validation or detailed material performance data [87,88]. Table 5 summarizes the principal advantages and limitations of microwave sintering-based additive manufacturing. This approach enables binder-free and water-free processing of regolith using electrically generated energy, independent of solar irradiance; however, challenges remain related to high energy demand, thermal gradient control, and variability in regolith composition.

Table 5
Summary of the main advantages and limitations of microwave sintering-based additive manufacturing for Martian construction, considering ISRU compatibility, energy requirements, process control, and material variability.

Aspect	Advantages	Disadvantages
ISRU compatibility	No binders or water required	Strong dependence on regolith composition
Heating efficiency	Volumetric heating	Difficult thermal control
Structural application	Suitable for shielding and infrastructure	Limited architectural precision
Energy demand	Electrically driven (can use solar power)	High energy consumption
Reliability	Works in enclosed or buried systems	Risk of microcracking

4.5. Laser sintering: printing small-scale components

Although laser sintering is not expected to be suitable for the construction of large-scale habitats, due to its inherently high precision and limited build rate, characteristics analogous to powder bed fusion processes, it remains a valuable manufacturing route for small-scale components on Mars. Fig. 7 illustrates the 3D printing process based on laser sintering carried out inside a Martian habitat.

Laser sintering can effectively process fine regolith powders to fabricate geometrically complex, high-resolution elements such as mechanical parts, tools, connectors, brackets, seals, replacement components required for habitat operation and maintenance. Its ability to produce near-net-shape parts with tight dimensional tolerances makes it particularly attractive for in-habitat manufacturing, where controlled environments and smaller build volumes are feasible. Consequently, laser sintering should be regarded not as a primary construction technology for Martian habitats, but as a complementary fabrication method supporting in situ production of precision components using locally available regolith-derived feedstocks. Table 6 summarizes the principal advantages and limitations of laser sintering-based additive manufacturing.

4.6. Robotic systems and instrumentation for additive manufacturing on Mars

The implementation of additive manufacturing for Martian

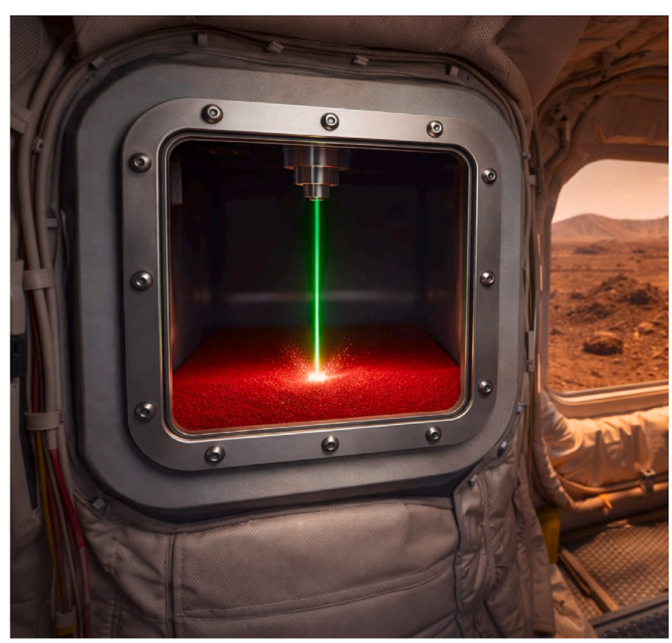

Fig. 7. Conceptual illustration of laser sintering-based 3D printing inside a Martian habitat, where fine regolith-derived powders are processed to fabricate small-scale, high-precision components for in situ manufacturing and maintenance.

Table 6
Summary of the main advantages and limitations of laser sintering-based additive manufacturing for Martian applications, considering precision, scalability, environmental requirements, and suitability for in-situ resource utilization.

Aspect	Advantages	Disadvantages
Precision	High-resolution fabrication	Very low build rate
Application	Ideal for tools and components	Not suitable for large structures
Environment	Works inside controlled habitats	Requires pressurized environment
Material use	Uses fine regolith powders	Powder handling complexity

construction extends beyond material formulation and printing techniques, requiring the integration of robotic systems, energy delivery units, and process control instrumentation. These components collectively enable excavation, material handling, deposition or sintering, and quality assurance under highly constrained environmental conditions. While the specific configuration of such systems will depend on the selected construction approach, several key categories of equipment can be identified as central to current technological concepts.

Robotic platforms constitute the backbone of autonomous construction systems. Mobile units, such as rover-based systems, are expected to perform site preparation, regolith excavation, and material transport, while stationary or semi-stationary systems, including gantry-based printers or robotic arms, enable the precise deposition or consolidation of materials. Extrusion-based processes require specialized printheads capable of handling regolith-based pastes or thermoplastic mixtures, often incorporating integrated heating and mixing systems. In contrast, sintering-based approaches rely on energy delivery systems such as solar concentrators, microwave generators, or laser sources, each presenting distinct operational constraints related to energy efficiency, controllability, and environmental compatibility.

Supporting instrumentation plays a critical role in ensuring process reliability and material performance. This includes thermal control systems for regulating curing or solidification, as well as sensors for monitoring temperature, material flow, layer geometry, and structural integrity. The integration of these sensing systems with advanced control algorithms, including machine learning-based approaches, is expected to be essential for adaptive process control under variable Martian conditions.

Table 7 summarizes the main robotic systems and instrumentation relevant to additive manufacturing for Martian construction, including their functions, advantages, limitations, and technological maturity. It should be noted that the systems discussed here represent the principal categories of equipment currently considered in the literature, rather than an exhaustive inventory of all possible technologies. A detailed analysis of robotic architectures, instrumentation design, and control strategies lies beyond the scope of the present review.

4.7. Energy requirements of additive manufacturing technologies

Energy demand represents a decisive factor in the selection of additive manufacturing technologies for extraterrestrial construction, particularly when comparing low-temperature extrusion-based processes with high-temperature thermal sintering routes. Extrusion-based systems, including polymer- and geopolymer-based material deposition, generally operate at near-ambient or moderately elevated temperatures and therefore exhibit relatively low direct process energy requirements. In these systems, energy consumption is dominated by mechanical operations such as mixing, pumping, and extrusion, rather than thermally driven phase transformations. Consequently, the printing stage itself is inherently energy-efficient, typically requiring on the order of ~0.36-1.8 MJ kg^{-1} for extrusion-based additive manufacturing processes [89].

However, the total energy demand of extrusion-based approaches is strongly influenced by upstream material processing. In the case of geopolymer-based systems, precursor preparation - including crushing, grinding, and chemical activation - constitutes a major energy hotspot, involving significant mechanical and chemical energy inputs [90]. This highlights that, although the deposition process itself is low-energy, the overall system demand can increase substantially when full life-cycle considerations are considered.

In contrast, thermal processing routes such as solar, laser, and microwave sintering require the regolith to be heated to high temperatures, typically exceeding 1000-1200 °C, to enable densification through melting or solid-state diffusion [89]. The energy requirement for these processes is fundamentally governed by the need to raise the material to sintering or melting temperatures and maintain these conditions over time. Thermodynamic analyses indicate that even under idealized assumptions, substantial energy input is required solely for heating, with additional increases when realistic inefficiencies are considered [91].

Experimental studies further confirm the high energy intensity of thermal sintering. Microwave-assisted sintering of lunar regolith simulants has been reported to require approximately ~69-98 MJ kg^{-1}, even under controlled laboratory conditions [92]. Although microwave processing is comparatively more efficient than surface-heating methods due to volumetric energy absorption and reduced heat losses [80], these values still illustrate the inherently energy-intensive nature of

Table 7
Robotic systems and instrumentation for additive manufacturing on Mars.

System/Component	Function	Application in AM	Advantages	Limitations	Technology readiness
Autonomous mobile robots (rovers)	Transport materials, site prep	Regolith excavation & delivery	High mobility, multi-purpose	Limited payload, energy constraints	High (space heritage)
Robotic arms (multi-DOF manipulators)	Material placement, tool handling	Extrusion/sintering heads	High precision, flexibility	Limited reach, slower build rates	High
Gantry-based systems	Large-scale printing	Habitat-scale extrusion	High stability, scalability	Deployment complexity	Medium
Extrusion printheads (DIW)	Deposit regolith-based paste	Geopolymer printing	Mature, scalable	Requires binder/water	Medium-high
Thermal extrusion systems (sulfur)	Melt & deposit sulfur-regolith mix	Thermoplastic printing	Water-free, fast solidification	Requires heating control	Medium
Solar concentrators (Fresnel lenses/mirrors)	Focus solar energy	Solar sintering	Renewable energy	Dependent on sunlight	Medium
Microwave generators	Volumetric heating	Microwave sintering	Works without sunlight	High energy demand	Low-medium
Laser systems	Localized heating	Powder bed sintering	High precision	Low scalability	High (small-scale)
Material handling systems	Sieving, transport, feeding	Feedstock preparation	Improves consistency	Additional complexity	Medium
Thermal control systems	Heating/cooling during processing	Curing & solidification	Improves material quality	Energy intensive	Medium
Sensors (temperature, rheology, imaging)	Process monitoring	Quality control	Enables automation	Requires robustness	Medium
AI/control systems	Process optimization	Autonomous printing	Adaptive, efficient	Requires validation	Emerging

high-temperature processing. Laser-based additive manufacturing processes exhibit lower but still significant energy demands, on the order of ~10.8-18 MJ kg^{-1}, reflecting the energy required to achieve localized melting at temperatures up to ~1600 °C [89]. In the case of solar sintering, energy input is delivered through highly concentrated solar fluxes (typically in the $MW·m^{-2}$ range), and although direct volumetric energy values are less frequently reported, the process is similarly constrained by high thermal requirements and significant radiative and conductive losses, resulting in comparable energy intensity to other high-temperature surface-heating techniques [80,89].

Additional insight is provided by studies on molten regolith processing, which show that maintaining regolith in a liquid state requires sustained temperatures above ~1200 °C, with continuous energy input necessary to counteract radiative and conductive losses and ensure process stability [93]. This further reinforces that manufacturing approaches relying on melting or high-temperature sintering are intrinsically constrained by both high initial heating demands and ongoing energy requirements during processing.

When compared on a consistent mass basis, extrusion-based additive manufacturing typically requires ~0.36-1.8 MJ kg^{-1} for the printing stage [89], increasing to approximately ~5-20 MJ kg^{-1} when upstream material preparation is included [90]. In contrast, thermal sintering processes generally range from ~10 to 20 MJ kg^{-1} for laser-based techniques [89] to ~70-100 MJ kg^{-1} for microwave sintering under experimental conditions [92], with solar sintering expected to fall within a similar range to other surface-heating methods depending on system efficiency and solar concentration [80]. This difference - spanning approximately one order of magnitude or more - is consistent with thermodynamic analyses showing that substantial energy input is required to raise regolith to sintering temperatures and maintain these conditions [91], reflecting the fundamental energy penalty associated with high-temperature processing.

This contrast highlights a critical trade-off in extraterrestrial construction strategies. Sintering-based methods enable binder-free fabrication and full utilization of in-situ materials, making them highly attractive for long-term infrastructure development. However, their high energy demand poses a significant limitation under the constrained power budgets of early lunar or Martian missions. Conversely, extrusion-based approaches offer a more energy-efficient pathway for initial construction, albeit with dependencies on binders and energy-intensive material preprocessing.

To facilitate a direct comparison of the principal material systems and additive manufacturing approaches discussed, Table 8 summarizes the key performance indicators, including mechanical properties, processing requirements, water consumption, and dependence on Earth-supplied resources.

5. Protective design and performance

Measurements performed by instruments operating on Mars, notably the Radiation Assessment Detector aboard National Aeronautics and Space Administration (NASA) rovers, indicate that the average radiation environment at the Martian surface is dominated by galactic cosmic rays (GCR), with an equivalent dose of approximately 0.64 $mSv·day^{-1}$ under steady-state conditions [94,95]. In addition to this persistent background, Solar Energetic Particles (SEPs) can produce sporadic but intense radiation events, delivering significantly elevated doses over short timescales and representing an important additional hazard for surface operations. For comparison, the average natural background radiation on Earth is about 2.4 $mSv·yr^{-1}$ [96], while astronauts aboard the International Space Station are exposed to approximately 0.5-1.0 $mSv·day^{-1}$ [97]. Career radiation dose limits vary among space agencies, typically ~0.6 Sv for NASA, ~1.0 Sv for Canadian Space Agency (CSA), European Space Agency (ESA) and Russian Space Agency (RSA), and ~0.5-1.0 Sv for Japan Aerospace Exploration Agency (JAXA), depending on age and sex [98]. Taking 1.0 Sv as a representative career dose limit, an astronaut continuously exposed to the average Martian surface radiation environment (~0.6-0.7 mSv day^{-1}) would reach this limit in approximately 4.0-4.6 years in the absence of adequate shielding. This estimate does not account for the cumulative radiation dose received during the interplanetary cruise phase, during which astronauts are exposed to dose rates of approximately 1.84 mSv day^{-1} in unshielded deep space [94].

It should be noted that the radiation environment on Mars is composed primarily of galactic cosmic rays (GCR) under steady-state conditions, while solar particle events (SPE) can produce significant short-term dose increases. Reported values in the literature depend on the adopted dose metric, typically expressed as equivalent dose ($mSv·day^{-1}$ or $mSv·yr^{-1}$) or absorbed dose ($mGy·yr^{-1}$), as well as on shielding assumptions and exposure geometry.

To address this issue, water-based shielding is preferentially concentrated around sleeping, eating, and working areas, where crew members spend most of their time. With such measures, effective daily doses within protected crew habitats could be reduced to ~0.5-1.0 mSv day^{-1} or lower, potentially keeping a full round-trip Mars mission (e.g., ~180-200 days of total interplanetary transit) below NASA's 600 mSv career dose limit for most astronauts, corresponding to an estimated ≤1% increase in lifetime fatal cancer risk [99]. According to NASA risk models [100], an absorbed dose of 1 Sv is associated with an approximate 3-7% increase in lifetime cancer risk for a middle-aged adult. Consequently, reducing interplanetary travel time has been extensively investigated as a critical mitigation strategy. For instance, Kingdon et al. [101] report the feasibility of a ~90-day Earth-Mars transit using a SpaceX Starship-class architecture, instead of the commonly reported

Table 8
Comparative summary of material systems for additive manufacturing on Mars [45–48,70,80,89,92].

Material system	Compressive strength (MPa)	Flexural strength (MPa)	Processing temperature (°C)	Water consumption	Earth-dependence	Notes
Geopolymer (extrusion)	~10–32	~5–9	~20–80 (curing)	High	High	Requires water and alkaline activators not readily available in-situ; strong dependence on imported chemical precursors
Sulfur-based (Martian Concrete)	~20–60	~9–13	~120–150	None	Low–Medium	Eliminates water use; sulfur potentially available on Mars, but processing and material handling may require additional infrastructure
Solar sintered regolith	~20–100 (process-dependent)	Limited data	>1000	None	Very Low	Fully based on in-situ regolith and solar energy; minimal need for imported materials
Microwave sintered regolith	~30–80 (process-dependent)	Limited data	>1000	None	Very Low	Utilizes regolith directly; dependence shifts to energy systems rather than material supply
Laser sintering (powder bed/directed energy)	~50–150 (localized)	Limited data	~1100–1600	None	Medium	Requires complex, high-precision equipment and controlled environments supplied from Earth

6-9 months [102].

Continuous radiation exposure on the Martian surface is another widely discussed challenge for long-term human presence. Using GEANT4 Monte Carlo simulations, Morthekai et al. [103] showed that the Martian surface receives an annual radiation dose of approximately 63 mGy·yr^{-1}, with a secondary-particle-induced dose maximum occurring at a depth of ~40 cm in the regolith, followed by a rapid attenuation such that radiation levels decrease to less than 2% of the surface value at depths greater than ~5 m, confirming Martian regolith as an effective radiation shielding material. These simulations assume basaltic regolith compositions with density values representative of loose to moderately compacted soil and a semi-infinite overburden geometry.

Llamas et al. [104] evaluated the effectiveness of Martian regolith as a radiation shield for human habitats, focusing on dose reduction as a function of burial depth under Martian surface conditions. Using analytical calculations based on particle stopping power and numerical simulations, the study showed that ~1 m of regolith can reduce primary radiation doses by approximately 40%, confirming the potential of regolith-covered or buried structures for radiation protection. However, the authors also highlighted that regolith alone is less effective in attenuating secondary radiation, suggesting that multilayer shielding systems, potentially combining regolith with hydrogen-rich materials, may be required for optimal protection. These results are generally associated with GCR-dominated radiation environments, as commonly considered in shielding simulations under steady-state Martian surface conditions, typically assuming simplified shielding geometries (e.g., planar or semi-infinite configurations).

Chen et al. [105] conducted a comprehensive modeling using Monte Carlo investigation of the Martian surface radiation environment using the Geant4 toolkit, with simulations benchmarked against MSL/RAD measurements. The study reports an average whole-body equivalent dose of ~153 mSv yr^{-1} for an unshielded human on the Martian surface, with significantly higher organ doses in the upper body, particularly the brain (~234 mSv yr^{-1}). Protons, alpha particles, and heavy ions were identified as the dominant contributors to the equivalent dose. Thus, the shielding effectiveness of Martian regolith was systematically evaluated by varying soil thickness from 0.1 to 3 m. Results indicate that ~1.0 m of regolith reduces the effective dose by ~71%, lowering the whole-body dose to ~35 mSv yr^{-1}, while 1.5 m reduces the dose to ~18 mSv yr^{-1}, below the ICRP occupational exposure limit (20 mSv yr^{-1}). At 3 m depth, the effective dose decreases further to ~2.7 mSv yr^{-1}, approaching terrestrial background radiation levels. These simulations typically assume isotropic radiation exposure and homogeneous regolith shielding, although specific assumptions may vary between studies. The main assumptions and modeling approaches reported in the literature are summarized in Table 9.

Early human habitation strategies on Mars increasingly favor hybrid construction concepts that combine lightweight, deployable pressurized modules with in situ resource utilization (ISRU) for environmental protection. Among these approaches, the integration of inflatable habitat modules, inspired by concepts such as NASA's TransHab [106, 107] and subsequent Bigelow Aerospace [108] developments, with regolith-based shielding has emerged as a particularly promising solution for initial surface installations. Martian regolith, a loose, granular layer covering most of the planet's surface, can be exploited as a multifunctional protective medium, providing effective attenuation against ionizing radiation, extreme diurnal temperature variations, dust-related hazards, and micrometeoroid impacts, while simultaneously minimizing the mass required to be launched from Earth. The operational simplicity of this concept is a major advantage during early mission phases. Inflatable habitats can be transported in a compact configuration, deployed and pressurized on the Martian surface, and subsequently covered with locally excavated regolith using robotic systems.

The choice of geometry is deliberate: domed and cylindrical shapes efficiently distribute internal pressure while reducing the surface-area-

Table 9
Summary of radiation environment and shielding performance assumptions for Martian habitats.

Study/Source	Radiation source	Dose metric	Shielding material	Thickness	Density assumption	Geometry/configuration	Model/method	Key result
MSL/RAD measurements [94,95]	GCR (steady-state)	Equivalent dose (mSv·day^{-1})	None (direct surface exposure with atmospheric shielding)	0 m (no additional shielding layer)	-	Surface measurement (hemispherical exposure due to planetary shielding)	In-situ measurement (RAD instrument)	~0.64 mSv·day^{-1} (surface dose equivalent)
Deep space transit [72]	GCR + SPE	Equivalent dose (mSv·day^{-1})	Spacecraft structural shielding (MSL capsule)	Not specified (spacecraft structural shielding)	-	Spacecraft interior (quasi-isotropic exposure within shielding structure)	In-situ measurement (RAD during interplanetary cruise)	~1.84 mSv·day^{-1} (cruise dose equivalent)
Morthekai et al. [103]	GCR + SPE (with secondary particle production)	Absorbed dose (mGy·yr^{-1})	Martian regolith	0-5 m (depth in regolith)	Typical regolith	Depth-dependent regolith shielding (semi-infinite medium approximation)	GEANT4 Monte Carlo radiation transport simulation	~63 mGy·yr^{-1} at surface; strong attenuation with depth
Llamas et al. [25]	GCR + SPE	Dose reduction (%)	Martian regolith	Depth-dependent (~1 m highlighted)	Typical regolith	Subsurface shielding (1D depth-dependent model)	Analytical calculations + SPENVIS (MEREM) modeling	~40% dose reduction at ~1 m depth (model-dependent)
Chen et al. [105]	GCR only	Equivalent dose (mSv·day^{-1})	Martian regolith	0.1-3 m	Typical regolith	Simplified surface geometry with hemispherical/isotropic incidence and regolith overburden	GEANT4 Monte Carlo simulation (validated against RAD measurements)	~71% reduction at 1 m; ~2.7 mSv·yr^{-1} at 3 m (GEANT4, GCR-only)

to-volume ratio, which in turn limits radiation exposure. Transparent silica aerogel panels, approximately 2 cm thick, can be used to create windowed areas while preserving high thermal insulation and attenuating radiation penetration [109]. Fig. 8 presents a schematic illustration of an inflatable habitat being buried beneath loose Martian regolith for environmental protection.

As an illustrative example, Roberts [110] proposed Kevlar-reinforced spherical habitat modules, in this case for lunar habitats, supported by a redundant metallic cage, providing approximately 40 m^3 of habitable volume, a total mass of about 2200 kg, and radiation protection afforded by a 3 m-thick regolith cover. Ortiz et al. [111] proposed a habitat concept designed to sustain a 1 m-thick layer of Martian regolith as external shielding, while employing hydrogen-rich materials, such as polyethylene, for enhanced internal radiation protection. The design further integrates a 25.4 cm-thick fiberglass layer, providing passive thermal insulation to mitigate external temperature fluctuations.

Internal operating pressures in inflatable habitats typically range from approximately 30 to 100 kPa, depending on life-support requirements. At these pressures, the internal load significantly exceeds the lithostatic stress imposed by several meters of Martian regolith under reduced gravity ($\sim$0.38 g), meaning that overburden does not govern structural design and burial depths of a few meters can be achieved without compromising structural integrity [112]. At these thicknesses, regolith shielding can significantly attenuate radiation exposure, with reductions commonly reported in the range of tens to over 80% depending on particle type and energy spectrum, while the overburden also acts as an effective thermal buffer, reducing diurnal temperature fluctuations and associated heating and cooling demands [105,112].

Despite these advantages, direct burial of inflatable habitats beneath loose regolith faces several limitations. Loose regolith is susceptible to differential settling induced by aeolian processes or seismic activity. Furthermore, fine dust particles may infiltrate interfaces and seals through electrostatic adhesion, increasing abrasion risks and maintenance demands.

To address these concerns, hybrid habitat concepts can be exploited by using consolidated regolith shelters between the inflatable core and the outer loose overburden. This layer may be produced through previous 3D printing processes discussed, forming a rigid or semi-rigid shell around the pressurized volume. Such a shell fundamentally transforms the structural behavior of the system by redistributing overburden loads more uniformly, mitigating regolith migration, and providing enhanced resistance to puncture and abrasion. In addition, consolidated regolith structures can be engineered with tailored density or graded porosity, improving shielding performance against high-energy particles and secondary radiation components, such as neutrons, which are less effectively attenuated by unconsolidated materials alone. Beyond structural and radiological benefits, the presence of a consolidated layer improves thermal stability by reducing direct regolith-membrane contact. Fig. 9 presents a visualization of the hybrid habitat with consolidated and loose regolith as protective shelters.

The principal trade-off associated with consolidated regolith shells lies in their energy and system requirements. Additive manufacturing processes demand power, time, and robotic complexity, which may be constrained during early surface missions. Consequently, loose regolith burial remains attractive for initial outposts prioritizing speed and simplicity. However, as in-situ manufacturing capabilities mature, hybrid systems increasingly represent a logical progression toward permanent habitation. Table 10 provides a comparative overview of the principal habitat design strategies for Mars, emphasizing the trade-offs between construction simplicity, energy demand, structural performance, and environmental protection. While loose regolith shielding enables rapid early deployment, more advanced concepts based on consolidated structures or subsurface construction offer superior long-term performance at the cost of increased complexity and resource requirements.

6. Structural response under Martian loading conditions

Beyond intrinsic material properties, the structural performance of Martian habitats must be evaluated under a unique multi-physics loading environment. This includes high-differential internal pressurization, long-duration seismic excitation, and soil-structure interaction (SSI) under a gravitational field of 3.71 m/s^2.

Seismic activity on Mars has been quantitatively characterized through data obtained by the InSight mission, which identified 465 seismic events within the first year of operation, with magnitudes ranging from Mw $\approx$ 1.3 to 3.7 [113]. These values are significantly lower than those associated with damaging earthquakes on Earth, where structural damage typically occurs for magnitudes above Mw $\approx$ 5.

In addition to their lower magnitude, marsquakes exhibit distinct characteristics in terms of signal duration, frequency content, and attenuation. Recorded events often last from several minutes to over 30 min, considerably longer than typical terrestrial earthquakes. Furthermore, seismic waves on Mars experience lower attenuation, allowing energy to propagate over longer distances and persist for extended periods [113,114].

Peak ground motion amplitudes recorded on Mars are on the order of nanometers, which is several orders of magnitude lower than those observed during terrestrial earthquakes, where displacements can reach millimeter to meter scale. As a result, seismic loading is not expected to govern ultimate structural strength design for Martian habitats. However, despite their low amplitude, marsquakes may still have important

Fig. 8. Conceptual view of an inflatable habitat being buried by loose Martian regolith for protection.

Fig. 9. Conceptual visualization of an inflatable habitat with a consolidated shelter and loose regolith for additional environmental protection.

Table 10
Comparison of main Martian habitat design approaches, summarizing their advantages, limitations, and associated trade-offs in terms of construction complexity, radiation protection, structural performance, and operational feasibility.

Design approach	Advantages	Limitations	Trade-offs
Loose regolith shielding (buried inflatable habitats)	Simple and rapid deployment; minimal infrastructure; excellent radiation shielding with sufficient thickness; low initial energy demand	Susceptible to regolith settling and dust infiltration; limited structural support; potential long-term stability issues	Favors speed and simplicity over durability and structural robustness
Hybrid systems (inflatable + consolidated regolith shell + loose overburden)	Improved structural stability; enhanced radiation and thermal performance; reduced regolith migration; better durability	Requires additive manufacturing capabilities; higher energy and operational complexity; slower deployment	Balances performance and protection with increased system complexity and energy demand
Fully consolidated regolith structures (3D printed/ sintered habitats)	High structural integrity; long-term durability; integrated architectural design; reduced reliance on inflatable systems	Challenges with airtightness and permeability; complex construction; high energy and processing requirements	Prioritizes long-term sustainability and structural performance at the expense of simplicity and early feasibility
Subsurface habitats (lava tubes/ excavated structures)	Excellent radiation shielding; stable thermal environment; minimal external exposure	Site-dependent availability; excavation complexity; limited geometric control	Maximizes environmental protection but reduces flexibility and increases site constraints

implications for structural performance due to their long duration and frequency characteristics. In particular, persistent low-amplitude excitation may induce resonance effects in lightweight or thin-walled structures, especially those produced by additive manufacturing. This is further influenced by the mechanical behavior of Martian regolith, which is expected to exhibit low damping and may amplify seismic waves, similarly to soft soils on Earth [113].

Overall, while seismic loads on Mars are significantly weaker than those on Earth, their characteristics suggest that structural design should consider dynamic response, resonance effects, and long-term fatigue rather than extreme load resistance. These aspects are particularly relevant for large-span habitats and additively manufactured structures, where material anisotropy and reduced damping may amplify dynamic effects. A comparative summary of key seismic characteristics and their structural implications is provided in Table 11.

Internal pressurization represents a primary load case for human habitats. Typical habitat pressures (on the order of 50-100 kPa) induce sustained tensile stresses in structural shells, while repeated pressurization cycles may lead to fatigue damage over time. This is especially relevant for additively manufactured materials, which may exhibit anisotropy, layer interfaces, and microstructural heterogeneities affecting crack initiation and propagation. Therefore, fatigue resistance and long-term durability must be considered alongside static strength.

Furthermore, soil-structure interaction is heavily influenced by the reduced gravity (0.38 g). While lower gravity reduces the overall dead load, it simultaneously decreases the effective confining pressure of the regolith, thereby altering its bearing capacity and shear strength. This makes large-span habitats particularly susceptible to differential settlement, especially when constructed on the loosely consolidated or layered deposits typical of automated ISRU construction. To the authors' knowledge, no studies have yet performed integrated simulations combining marsquake ground motion, regolith mechanics, and the structural response of additively manufactured habitats. This represents a critical gap for future design-oriented research, as structural challenges on Mars extend beyond material selection to require the integrated consideration of loading conditions, construction methods, and long-term performance.

7. Final remarks

The future development of 3D-printed habitats on Mars will depend on the convergence of material science, autonomous construction technologies, and energy-efficient processing systems. Rather than a single dominant solution, progress is expected to follow a staged and hybridized approach, in which different additive manufacturing techniques are deployed according to mission phase, environmental constraints, and infrastructure maturity.

In this context, 3D printing should be viewed not as a replacement for alternative construction strategies, but as a complementary technology that becomes increasingly dominant as ISRU capabilities, energy availability, and robotic autonomy evolve. Although additive manufacturing enables the in-situ production of structural elements using Martian regolith, it does not eliminate the need for Earth-supplied components. Critical subsystems, including life-support equipment, sealing materials, mechanical interfaces, electronics, and high-precision components, are unlikely to be manufactured directly from regolith in early mission phases. Consequently, 3D printing should be understood as a strategy for reducing, rather than eliminating, dependence on transported materials.

In the near term, the primary challenge lies in the automation and reliability of construction systems. Robotic platforms must be capable of

Table 11
Comparison between Martian and terrestrial seismic loading and implications for structural design [105–121].

Parameter	Mars	Earth	Structural implication
Magnitude range (Mw)	~1.3 - 4.2	0 - 9+	Marsquakes are too weak to cause structural collapse; not governing ultimate design
Typical damaging events	Not observed	Mw > 5	Structural safety on Mars governed by other loads (e.g., internal pressure)
Ground displacement	~10^{-11} - 10^{-9} m	~10^{-6} - 10^{-1} m	Extremely low deformation; negligible direct stress
Signal duration	Seconds to several minutes (with long codas and extended signal durations)	Seconds to minutes	Long-duration excitation may influence dynamic response
Frequency content	~0.1 - 10 Hz	~0.1 - 20 Hz	Potential overlap with structural natural frequencies
Attenuation	Low apparent attenuation inferred from long-duration signals and persistent codas	Higher attenuation	Persistent vibrations over time
Event frequency	300-500 events/year (low magnitude)	Variable	Repeated loading may contribute to fatigue
Soil behavior	Loose, low damping regolith	Variable soils	Possible amplification and site effects
Dominant structural effect	Dynamic serviceability, fatigue	Strength and collapse (in strong events)	Different design philosophy required

operating with minimal human intervention, performing excavation, material handling, and fabrication under highly variable environmental conditions. This requires advances in autonomous navigation, real-time process monitoring, and adaptive control systems capable of compensating for variability in regolith composition, temperature, and atmospheric conditions. The integration of sensing technologies with machine learning algorithms for defect detection and process optimization represents a critical area for future research.

From a materials perspective, a key priority is the development of fully ISRU-compatible formulations that minimize or eliminate dependence on Earth-supplied additives. For extrusion-based systems, this implies the search for alternative binders derived entirely from Martian resources, such as sulfur-modified or hybrid inorganic-organic systems. For sintering-based approaches, further work is required to optimize particle size distribution, phase composition, and thermal processing conditions to achieve consistent densification and mechanical performance across different regolith types. In particular, controlling porosity and permeability remains essential for ensuring structural integrity and, where required, gas tightness.

Energy efficiency will play a defining role in the scalability of additive manufacturing on Mars. Future systems must balance power consumption, construction speed, and material performance, especially under constraints imposed by solar intermittency and limited energy storage capacity. Hybrid energy strategies, combining solar concentration, microwave heating, and resistive systems, may offer flexible solutions adapted to local conditions and mission timelines. The development of low-energy processing routes, including rapid sintering techniques and optimized thermal cycles, is therefore a key research direction.

Another critical aspect concerns the integration of structural and functional design. Future habitats will likely incorporate multi-material and functionally graded structures, enabling simultaneous optimization of mechanical strength, thermal insulation, and radiation shielding. Additive manufacturing offers unique opportunities in this regard, particularly through the controlled variation of density and microstructure at different length scales. The combination of printed structural shells with loose or partially consolidated regolith layers represents a promising pathway toward high-performance hybrid systems.

At the architectural level, future developments should move beyond purely structural considerations toward system-level integration, including life-support interfaces, airtightness solutions, and repairability. One of the main unresolved challenges is the reliable sealing of 3D-printed regolith structures under internal pressurization. This will likely require the incorporation of internal liners, coatings, or composite layers, as well as standardized interfaces for airlocks and modular expansion.

Looking further ahead, the long-term sustainability of Martian settlements will depend on the scalability and adaptability of construction technologies. This includes the ability to reuse and recycle materials, repair damaged structures, and expand infrastructure incrementally. Technologies such as remelting of sulfur-based materials, re-sintering of regolith, and in-situ fabrication of replacement components will be essential for reducing logistical dependence on Earth.

Future research should adopt a more integrated, system-oriented perspective, linking material development, processing technologies, energy systems, and habitat design within a unified framework. Experimental validation under Mars-analog conditions, combined with numerical modeling and mission-scale simulations, will be essential to bridge the gap between laboratory demonstrations and real-world deployment.

In this context, 3D printing should be viewed not as a replacement for alternative construction strategies, but as a complementary technology whose relevance increases with the maturation of ISRU capabilities, energy availability, and robotic autonomy. However, even in advanced scenarios, fully autonomous in-situ fabrication of complete habitats remains unlikely. Instead, future systems are expected to rely on hybrid approaches that combine regolith-based structural printing with the integration of prefabricated components delivered from Earth or assembled in orbit.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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